Agilex 5 Camera 4Kp60 HDR — Lab Bring-up Guide

Field	Value
Design	4Kp60 Multi-Sensor HDR Camera Solution System Example Design
Release	Use a matching pre-built QSPI + microSD pair; 25.1 and 26.1 are both documented
Hardware	Agilex 5 FPGA E-Series **065B Modular Development Kit**
Cameras	Framos FSM:GO **IMX678** (1 or 2 modules)
Audience	Lab / engineering team
Doc status	Team bring-up procedure with Ethernet validation and board-triage guidance

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1. Purpose

This document tells a teammate how to **boot and run** the pre-built Altera camera design on the Modular Development Kit, open the web GUI, and get camera video on DisplayPort.

It has two layers:

1. **Official Altera flow** (from the design documentation)
2. **Lab-specific notes** discovered during bring-up, including Ethernet validation and board triage

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2. Customer demo fast start (pre-provisioned kit)

Use this section when the kit has already been prepared and you need to bring it up in front of a customer. It is deliberately short; use the detailed procedure in Section 7 if a step fails.

What must already be ready

Item	Required state
QSPI	`top.core.jic` was programmed once successfully
microSD card	Contains the matching `hps-first-vvp-isp-demo-image-agilex5_mk_a5e065bb32aes1.wic` image
Boot switches	SOM SW1 is **ON-ON** (QSPI / AS×4 boot)
Camera	Framos IMX678 connected to MIPI0 or MIPI1, with pin 1 aligned
Display	Carrier **DisplayPort TX** connected to the selected monitor

Item	Required state
Host connections	Board J6 Ethernet and SOM J2 USB connected to the host PC
Host network	Board and host are on the same Ethernet network; VPN is disabled if it interferes

> A prepared microSD card alone is not enough. QSPI must already contain the matching `top.core.jic` boot image.

Demo sequence

1. Power on
Verify the connections above, insert the microSD card, and power on the board.

2. Wait for application startup
Wait about two minutes for Linux and the camera application to start.

3. Open the HPS serial console
Use 115200 8N1, press Enter, and log in as `root`:

```
sudo minicom -D /dev/ttyUSB6 -b 115200
```

`ttyUSB6` is the usual lab mapping; if it is blank, use the serial-port guidance in Section 7.3.

4. Get the board address

```
ip -4 addr show eth0
```

5. Open the GUI on the host PC

```
http://<board-ip>/
```

6. Select the camera input
In the web GUI, select **Input Config → Input Source → Camera**.

7. Confirm video
Confirm camera video appears on the DisplayPort monitor.

If the GUI does not open

First test normal Ethernet operation:

```
ping -c 2 -M do -s 1472 <board-ip>
```

- If this succeeds, keep the default MTU 1500 and continue with the GUI.
- If this fails, **do not lower MTU**. Follow the Ethernet validation and board-triage process in Section 8 before using the kit for a customer demonstration.

If the monitor has no video

In the GUI select **Input TPG** first. If the test pattern appears, switch back to **Camera**. For the MIPI1 single-camera known issue, toggle **Camera → Input TPG → Camera**.

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3. Official references

Resource	Link
Design overview	https://altera-fpga.github.io/rel-26.1/embedded-designs/agilex-5/e-series/modular/camera/camera_4k/camera_4k/
Detailed guide	https://github.com/altera-fpga/agilex5-ed-camera/blob/rel/26.1/docs/es/camera/camera_4k/camera_4k.md
Repository	https://github.com/altera-fpga/agilex5-ed-camera/tree/rel/26.1
Pre-built binaries	https://github.com/altera-fpga/agilex5-ed-camera/releases/tag/rel-26.1-isp_hdr-MDK_RevB_GrpB
25.1 test documentation	https://altera-fpga.github.io/rel-25.3/embedded-designs/agilex-5/e-series/modular/camera/camera_4k/camera_4k/
25.1 pre-built binaries	https://github.com/altera-fpga/agilex5-ed-camera/releases/tag/rel-25.1
Repo quick start	https://github.com/altera-fpga/agilex5-ed-camera/blob/rel/26.1/AGX_5E_Altera_Modular_Dk_ISP_designs/HDR_CAMERA.md
Known issues	https://github.com/altera-fpga/agilex5-ed-camera/blob/rel/26.1/docs/es/camera/camera_4k/known_issues.md

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4. Does the official design require setting MTU?

No. Altera's official camera 4K documentation does **not** ask you to change Ethernet MTU.

Official network guidance is only:

- Connect board **J6** Ethernet to the same network as the host PC (direct cable, switch, or router)
- Disable VPN on the host if it interferes
- Get the board IP with `ifconfig` / `ip a` after login as `root`
- Open `http://<board-ip>/` in a browser

Lab conclusion: keep MTU at 1500

On a **healthy** Ethernet path (normal 1500-byte frames work):

- You do **nothing** with MTU
- GUI and board work at default MTU 1500

During the initial lab bring-up, one board showed a size-dependent Ethernet failure:

- Frames larger than roughly 400 bytes were dropped
- Symptoms at MTU 1500 were:
- `ping` small packets OK
- `ping -M do -s 1472` fails
- Browser page hangs / blank
- `curl` may show `200 OK` but receive **0 bytes** of the HTML body before timing out
- SSH key exchange can stall

Using MTU 400 only hid the fault: 400-byte packets were still intermittently lost and the GUI greyed out. It is **not** an approved team workaround.

The same 25.1 image, host PC, and default MTU 1500 passed the full-size ping and GUI download after replacing the board. This isolates the incident to the original board's J6/HPS Ethernet path or associated board hardware; it is not an Ubuntu, browser, IP-version, or camera-design release issue.

Team rule: Keep MTU at 1500. First test full-size ping. If it fails, quarantine the board for Ethernet triage rather than reducing MTU.

```
# Replace BOARD_IP with the board address
ping -c 2 -M do -s 1472 BOARD_IP
```

Result	Action
Large ping succeeds	Keep default MTU 1500 (official / preferred)
Large ping fails	Follow Ethernet validation and board triage (Section 8); do not lower MTU

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5. One-time programming (select a matching release pair)

Repeat only if microSD or QSPI is wiped.

Release pairing rule

Never mix a QSPI `.jic` file from one release with a microSD `.wic` image from another release. Keep each pair in a separate host directory.

Release	Matching source	Lab status
25.1	Release tag `rel-25.1`	Validated on replacement board at MTU 1500
26.1	Release tag `rel-26.1-isp_hdr-MDK_RevB_GrpB`	Current documented release; do not pair with 25.1 assets

Both release pairs use these filenames:

File	Purpose
`top.core.jic`	QSPI boot image
`hps-first-vvp-isp-demo-image-agilex5_mk_a5e065bb32aes1.wic.gz`	microSD Linux + app image

Burn microSD

```
gunzip -k hps-first-vvp-isp-demo-image-agilex5_mk_a5e065bb32aes1.wic.gz
# Identify the whole-disk device carefully (e.g. /dev/sdX), not a partition
sudo dd if=hps-first-vvp-isp-demo-image-agilex5_mk_a5e065bb32aes1.wic \
  of=/dev/sdX bs=1M status=progress conv=fsync
sync
sudo eject /dev/sdX
```

Insert the card in the **SOM** microSD slot.

Program QSPI once

1. Power **off**
2. SOM **SW1 = OFF-OFF** (MSEL = JTAG)
3. Power **on**
4. Program:

```
jtagconfig
quartus_pgm -c 1 -m jtag -o "pvi;top.core.jic"
```

1. Power **off**
2. SOM **SW1 = ON-ON** (MSEL = AS×4 / QSPI boot)

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6. Hardware checklist (every session)

Item	Setting / connection
Boot mode	SOM **SW1 = ON-ON**
microSD	Inserted in **SOM** slot
Ethernet	Board **J6** → host PC Ethernet (lab preferred: direct shared link)
USB	SOM **J2** + Carrier **J35** both to PC
Cameras	Framos IMX678 on MIPI0 and/or MIPI1, pin 1 aligned
Display	Carrier **DisplayPort TX** → 4K-capable monitor (prefer native DP cable)

> USB serial devices (`/dev/ttyUSB*`) appear only when the board is powered on.

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7. Every-boot procedure (lab recommended)

7.1 Boot the host PC first

Configure Wired Ethernet as **Shared to other computers**.

```
ip -4 addr show enp3s0
# Expected on this lab PC: inet 10.42.0.1/24
```

If sharing is not active (connection name on this PC is `netplan-enp3s0`):

```
sudo nmcli connection modify "netplan-enp3s0" ipv4.method shared
sudo nmcli connection up "netplan-enp3s0"
ip -4 addr show enp3s0
```

7.2 Power on the board

Wait about **2 minutes** for Linux and `VvpIspDemo` to start.

7.3 Open HPS serial console

With **J2 + J35** connected you typically get **8** USB serial ports.

```
ls -l /dev/ttyUSB*

gnome-terminal --title="HPS" -- bash -c \
  "sudo minicom -D /dev/ttyUSB6 -b 115200; exec bash"
```

- Settings: **115200 8N1**, hardware flow control **OFF**
- Press **Enter**
- Login: `root` (no password)

Serial port map (important)

Port group (typical)	Role
`ttyUSB0` – `ttyUSB3`	JTAG / Nios group
`ttyUSB2`	Often **Nios** (“WELCOME TO INTEL NIOS SYSTEM”) — not Linux
`ttyUSB4` – `ttyUSB7`	HPS UART group
`ttyUSB6`	Often **HPS Linux** (3rd port of HPS group)

`ttyUSB` numbers can change after reboot. If `ttyUSB6` is blank, try `ttyUSB4` / `5` / `7`, or open all four and power-cycle once with minicom already open.

7.4 Read board IP

```
ip -4 addr show eth0
# Typical on shared link: 10.42.0.212 (can change)
```

If no IPv4 address:

```
udhcpc -i eth0
ip -4 addr show eth0
```

> Always re-read the IP. Do not assume it stays `.212`.

7.5 Network health check (validate Ethernet)

On the host:

```
BOARD_IP=10.42.0.212   # replace with actual eth0 IP

ping -c 2 $BOARD_IP
ping -c 2 -M do -s 1472 $BOARD_IP
```

- If **1472 succeeds** → continue at MTU 1500
- If **1472 fails** → stop GUI testing and follow Section 8; do **not** lower MTU

7.6 Open the web GUI

```
curl -sS --max-time 10 http://$BOARD_IP/ | wc -c
# Healthy response: nonzero HTML (about 1035 bytes for 25.1; about 1112 bytes for 26.1)

google-chrome-stable http://$BOARD_IP/
```

You should see the Altera splash, then the camera GUI.

7.7 Select camera input

In the GUI:

1. Open **Input Config**
2. Set **Input Source → Camera**
3. If video is missing (especially single camera on MIPI1): toggle **Camera → Input TPG → Camera** (official known-issue workaround)

Optional board log check:

```
grep -iE 'Found camera|Active input|GMSL' /tmp/vvp-isp-demo.log | head -n 30
```

Healthy example:

- `Found camera: FSM-IMX678` as Cam 0 / Cam 1
- `Active input: Camera 0`
- `3840x2160p @60`

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8. Ethernet validation and board triage

Use this section if small pings work but `ping -M do -s 1472` fails, or if the GUI is blank despite a working camera/DisplayPort pipeline.

Required normal configuration

- Board `eth0`: MTU **1500**
- Host Ethernet: MTU **1500**
- Host and board: same IPv4 network
- Host VPN/proxy: disabled if it interferes

1. Confirm the board application works locally

```
wget -S -O /tmp/gui-local.html -T 5 http://127.0.0.1/
wc -c /tmp/gui-local.html
```

If this returns nonzero HTML but the host browser is blank, continue below.

2. Validate the host-to-board Ethernet path

```
ping -c 2 $BOARD_IP
ping -c 2 -M do -s 1472 $BOARD_IP
curl -sS --max-time 10 http://$BOARD_IP/ | wc -c
```

Expected result: the full-size ping succeeds and `curl` receives nonzero HTML.

3. If full-size traffic fails

1. Keep MTU 1500; do not use a smaller MTU as a permanent workaround.
2. Replace the Ethernet cable with a known-good Cat5e/Cat6 cable.
3. Test directly against a second host PC with a normal 1500-MTU Ethernet connection.
4. If the failure follows the board, remove it from customer/demo use and open a board support case.

Known lab incident and resolution

- **Original board:** Failed full-size IPv4 and IPv6 traffic on Ubuntu and Windows, directly and through different network paths. Camera output and local web GUI were working. Packet loss began near 400 bytes.
- **Replacement board:** Using the same 25.1 QSPI/microSD pair and Ubuntu host, full-size ping passed and the web GUI returned a 1035-byte HTML response at MTU 1500.

The evidence identifies the original board's Ethernet/J6/HPS network path as faulty, though it does not identify the exact component.

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9. DisplayPort blank / “please load image”

Cameras can be detected in software while the monitor stays blank. That is usually a **DisplayPort link** issue, not a sensor detect failure.

1. In GUI, set **Input Source = Input TPG** - If TPG appears → DP works; switch back to **Camera** - If still blank → check cable / TX connector / monitor input
2. Prefer native **DP→DP**; cheap DP→HDMI adapters often fail at 4Kp60
3. Use carrier **DisplayPort TX**
4. Reseat cable; power-cycle monitor
5. Optional: watch Nios console (`ttyUSB2`) while replugging DP for link messages

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10. Useful board commands

```
ps | grep VvpIspDemo
tail -n 50 /tmp/vvp-isp-demo.log

# Confirm embedded web server responds locally
wget -S -O /tmp/local.html -T 5 http://127.0.0.1/
wc -c /tmp/local.html

# Restart application if needed
killall VvpIspDemo
cd /home/root && ./start.sh
```

App location: `/home/root/VvpIspDemo` (started by `/home/root/start.sh`).

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11. What not to redo every boot

- Do **not** rewrite the microSD image
- Do **not** reprogram QSPI (`top.core.jic`) unless flash was erased or boot is broken

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12. Troubleshooting quick table

Symptom	Likely cause	Action
No `/dev/ttyUSB*`	Board off / USB unplugged	Power on; connect J2 and J35
Only 4 ttyUSB ports	One USB cable missing	Connect both J2 and J35
Only Nios banner	Wrong serial port	Use HPS group’s 3rd port (often `ttyUSB6`)

Symptom	Likely cause	Action
Small ping works but browser blank / curl body = 0	Large-frame Ethernet failure	Run the 1472 ping test and follow Section 8; do not lower MTU
IP not `.212`	DHCP lease changed	Re-read `ip -4 addr show eth0`
Cameras found, monitor blank	DP link	Try Input TPG; check DP cable/TX
GUI greys out every few seconds	Intermittent Ethernet transport fault	Validate full-size ping at MTU 1500; compare with a known-good board
VPN connected	Host network interference	Disable VPN (official guidance)

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13. One-page cheat sheet

```
1. PC Wired = Shared                -> 10.42.0.1
2. Power board                      -> SW1 ON-ON
3. minicom /dev/ttyUSB6 115200      -> root
4. Board: ip -4 addr show eth0
5. Host:  ping -M do -s 1472 BOARD_IP
   - fail  -> do not lower MTU; follow board Ethernet triage
   - pass  -> keep MTU 1500 and open GUI
6. Chrome http://BOARD_IP/
7. GUI Input Config -> Camera
   (if needed: Camera <-> TPG <-> Camera)
```

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14. Document history

Version	Notes
1.0	Initial lab bring-up capture
1.1	Reformatted for team use; clarified that MTU is **not** an official Altera requirement and is a lab-path workaround only
1.2	Added customer demo fast-start section for a pre-provisioned board and microSD card
1.3	Replaced the MTU-400 recommendation with validated Ethernet triage after a replacement board passed full-size traffic and stable GUI operation at MTU 1500